

Ivan has given you N boxes. Initially, the i -th box has A_i marbles. You can move marbles around. In one operation, you can choose a marble in box i (where box i has at least one marble of course) and transfer it to either box $i - 1$ or box $i + 1$ (if it exists).

Ivan is very particular with the number of marbles in each box; he wants you to make some number of operations such that the i -th box ends up with B_i marbles. What is the minimum number of operations needed to satisfy Ivan?

It is guaranteed that $A_1 + A_2 + \dots + A_N = B_1 + B_2 + \dots + B_N$.

Input Format

The input consists of an integer N and a two arrays of integers A and B , each of length N .

Constraints:

- $1 \leq N \leq 5 \cdot 10^4$.
- $A_1 + A_2 + \dots + A_N \leq 5 \cdot 10^{11}$.
- $A_1 + A_2 + \dots + A_N = B_1 + B_2 + \dots + B_N$.

Output Format

Output a single integer, the minimum number of operations needed.

Sample Input

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N = 5
A = [2, 2, 2, 6, 3]
B = [1, 2, 3, 4, 5]
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Sample Output

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4
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Explanation

One possible optimal set of moves is:

1. Move one marble from box 1 to box 2. The marbles counts are now $[1, 3, 2, 6, 3]$.
2. Move one marble from box 2 to box 3: $[1, 2, 3, 6, 3]$.
3. Move one marble from box 4 to box 5: $[1, 2, 3, 5, 4]$.
4. Move one marble again from box 4 to box 5: $[1, 2, 3, 4, 5]$.

4 operations in total are used.

Ivan memberi anda N kotak. Pada awalnya, kotak ke- i mempunyai A_i guli. Anda boleh memindahkan guli-guli tersebut. Dalam satu operasi, anda boleh memilih satu guli dari kotak i (apabila kotak i mempunyai sekurang-kurangnya satu guli) dan memindahkannya sama ada ke kotak $i - 1$ atau kotak $i + 1$ (jika ia wujud).

Ivan amat teliti mengenai bilangan guli dalam setiap kotak; dia mahukan anda menjalankan beberapa operasi supaya kotak ke- i berakhir dengan B_i guli. Apakah bilangan operasi paling sedikit yang diperlukan bagi memenuhi kehendak Ivan?

Adalah dijamin bahawa $A_1 + A_2 + \dots + A_N = B_1 + B_2 + \dots + B_N$.

Format Input

Input terdiri daripada suatu integer N dan dua tatasusunan (array) integer A dan B dengan panjang N .

Kekangan:

- $1 \leq N \leq 5 \cdot 10^4$.
- $A_1 + A_2 + \dots + A_N \leq 5 \cdot 10^{11}$.
- $A_1 + A_2 + \dots + A_N = B_1 + B_2 + \dots + B_N$.

Format Output

Output satu integer, iaitu bilangan minimum bagi operasi yang diperlukan.

Contoh Input

$N = 5$

$A = [2, 2, 2, 6, 3]$

$B = [1, 2, 3, 4, 5]$

Contoh Output

4

Penjelasan

Satu turutan operasi minimum yang mungkin ialah:

1. Pindahkan satu guli dari kotak 1 ke kotak 2. Kiraan guli sekarang ialah $[1, 3, 2, 6, 3]$.
2. Pindahkan satu guli dari kotak 2 ke kotak 3: $[1, 2, 3, 6, 3]$.
3. Pindahkan satu guli dari kotak 4 ke kotak 5: $[1, 2, 3, 5, 4]$.
4. Pindahkan satu guli lagi dari kotak 4 ke kotak 5: $[1, 2, 3, 4, 5]$.

Sejumlah 4 operasi telah dibuat.

Solution

Solution 1

Let D be defined by $D_i = A_i - B_i$. Our problem can be reformulated as:

Let there be N boxes, where box i has $|D_i|$ marbles if D_i is positive and $|D_i|$ "holes" otherwise. We can move a marble **or a hole** to an adjacent box in one operation. If a marble meets a hole in the same box, they both disappear. What is the minimum number of operations needed to make all marbles and holes disappear? It is guaranteed that the number of marbles and holes are equal.

Note that moving a hole from box i to box j (through multiple operations) in the reformulated statement is equivalent to moving a marble from box j to box i in the original problem.

Consider box 1. There is only one optimal way to deal with it: move $|D_1|$ marbles or holes from box 1 to box 2, which takes $|D_1|$ operations. Once we have dealt with box 1, we now have the

exact same problem except without box 1! Thus, we can simply simulate dealing with the leftmost unprocessed box.

Let our answer initially be $ans = 0$ and the number of marbles/holes on the leftmost unprocessed box be $cur = 0$. For each i from 1 to N :

1. Add D_i to cur .
2. Add $|D_i|$ to ans .

ans will be our answer in the end.

Solution 2

Let A' and B' be defined by $A'_i = A_1 + A_2 + \dots + A_i$ and $B'_i = B_1 + B_2 + \dots + B_i$ for all integers i ($1 \leq i \leq N$).

Observe that:

1. Moving a marble from box i to $i - 1$ is equivalent to increasing A'_{i-1} by 1.
2. Moving a marble from box i to $i + 1$ is equivalent to decreasing A'_i by 1.
3. A and B are equal if and only if A' and B' are equal.

Now our problems reduces to “How many times do we need to increase or decrease some elements of A' by 1 to make it equal to B' ?” and this is very easy. The answer is simply the sum of $|A'_i - B'_i|$ over all integers i ($1 \leq i \leq N$)! Final tiny detail, A' and B' can be computed in $O(N)$ time just like how one computes a prefix sum.

Both solutions have time complexity $O(N)$. Furthermore, taking a closer look, solution 1 actually also counts the sum of $|A'_i - B'_i|$, only from a different perspective!